

Safety Data Sheet (SDS)

Section 1: Identification

Product identifier: Nital Etch, 2% to 5%

Other name(s): Nital, Nital, Nitrol

Item number(s): 101 to 107

Identified use: SU24 Scientific research and development

Details of the supplier of the safety data sheet:

ES Laboratory, LLC

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Emergency telephone number:

CHEMTREC® 1-800-424-9300 (US & Canada Only)

Section 2: Hazard(s) Identification

Hazardous classification of the substance or mixture:

Hazard Class	Category code
Flammable liquid	2
Skin Corrosion	1B
Serious Eye Damage	1
Acute toxicity (oral)	3
Acute toxicity (inhalation)	3
Specific Target Organ Toxicity – Single Exposure	1

Signal word: Danger

Pictogram:



Hazard statement(s):

- H225 Highly flammable liquid and vapor.
H314 Causes severe skin burns and eye damage.
H301+H331 Toxic if swallowed or inhaled.
H370 Causes damage to organs.

Precautionary statement(s):

- P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P260 Do not breathe dust/fume/gas/mist/vapors/spray.
P280 Wear protective gloves/protective clothing/eye protection/face protection.

Response statement(s):

- P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301+P330+P331+P310 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. Immediately call a POISON CENTER or doctor/physician.

Storage statement(s):

- P403+P233+P235 Store in a well-ventilated place. Keep container tightly closed. Keep cool.
P405 Store locked up.

Disposal statement(s):

- P501 Dispose of contents/container in accordance with local, regional, national, and international regulations.

Hazard(s) not otherwise classified: No information.

Label elements: See tables above

HMIS Ratings:

Health: 3

Flammability: 3

Reactivity: 1

NFPA Ratings:

Health: 3

Flammability: 3

Reactivity: 1

Special hazard: None

Section 3: Composition/Information on Ingredients

Component	CAS No.	Concentration (wt%)
Methanol (methyl alcohol)	67-56-1	86-91%
Nitric Acid (HNO ₃)	7697-37-2	2-6%

Any concentration shown as a range is to protect the confidentiality or is due to batch variation. Only hazardous components are shown.

Section 4: First-Aid Measures

Eye contact: Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Get medical aid immediately.

Inhalation: Remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical aid immediately.

Skin contact: Wash the areas of contact with water for at least 15 minutes while removing contaminated clothing and shoes. Get medical aid immediately.

Ingestion: Do not induce vomiting. Rinse mouth. Get medical aid immediately.

Most important symptoms and effects, both acute and delayed: May be fatal or cause blindness if swallowed. Impair liver or kidney function.

Recommendation for immediate medical care and special treatment needed, when necessary: No information.

Section 5: Fire-Fighting Measures

Extinguishing media: Water Spray, alcohol-resistant foam, dry chemical, or carbon dioxide.

Special hazards arising from the substance or mixture: In the case of fire, the following can be released: nitrogen oxides, carbon monoxide, and carbon dioxide.

Special protective equipment or precautions for firefighters: Wear full protective clothing and a self-contained respirator.

Section 6: Accidental Release Measures

Personal precautions, protective equipment and emergency procedures: Wear protective equipment. Keep unprotected persons away. Ensure adequate ventilation. Keep away from ignition sources.

Environmental precautions: Do not allow the material to be released into the environment without proper governmental permits.

Methods and materials for containment and cleaning up: Absorb with a liquid binding material (sand, diatomite, acid binder, universal binders, sawdust). Dispose of contaminated material as waste according to section 13. Ensure adequate ventilation. Keep away from ignition sources.

Section 7: Handling and Storage

Precautions for safe handling: Wear protective equipment. Ensure good ventilation in the workplace. Open and handle the container with care. Protect against electrostatic charges. Fumes can combine with air to form an explosive mixture.

Condition for safe storage: Keep the container tightly sealed. Store in an approved flammable liquid storage container/area.

Incompatibilities: Store away from oxidizing agents, strong bases, and reducing agents.

Specific storage requirement(s): No information.

Section 8: Exposure Controls/Personal Protection

Exposure Limits

Component	CAS No.	ACGIH TLV	OSHA PEL
Methanol (methyl alcohol)	67-56-1	200 ppm TWA skin 250 ppm STEL skin	200 ppm TWA
Nitric Acid	7697-37-2	2 ppm TWA 4 ppm STEL	2 ppm TWA

Engineering controls: Use general or local exhaust ventilation to control the vapor concentration.

Eye protection: Wear safety glasses or goggles.

Skin protection: Wear protective clothing and chemical-resistant gloves.

Respiratory protection: Use a self-contained respiratory device in an emergency situation.

Section 9: Physical and Chemical Properties

Appearance:	Clear liquid
UFL/UEL:	36 Vol % (methanol)
LFL/LEL:	6-6.7 Vol % (methanol)
Odor:	Alcohol-like
Vapor pressure:	96 mm Hg @20 °C (methanol)
Odor threshold:	Not determined
Vapor density:	~1.1 (air = 1, methanol)
pH:	Not applicable (non-aqueous)
Relative density:	0.791 g/cm ³ (methanol)
Melting Point/Freezing Point:	-98 °C (methanol)
Solubility in water:	Miscible
Boiling point/boiling range:	64.7 °C (methanol)
Flashpoint:	11 °C (methanol)
Evaporation Rate:	Not determined
Flammability (solid, gas):	Not applicable
Partition coefficient (n-octanol/water):	log Kow ~-0.77 (methanol)
Auto-ignition temperature:	~464 °C (methanol)
Decomposition temperature:	Not determined
Viscosity:	Not determined

Section 10: Stability and Reactivity

Reactivity: No information.

Chemical stability: Stable under recommended conditions.

Stabilizer(s): Not required.

Safety issues that may arise should the product change in appearance: No information.

Thermal decomposition/ conditions to avoid: Decomposition will not occur if used and stored according to specifications.

Possibility of hazardous reactions: see incompatibilities.

Incompatibilities: Strong oxidizer, reducing agent, heat, sparks, open flame. Will attack some forms of plastics, rubber, and coatings. May react with metallic aluminum and generate hydrogen gas.

Hazardous decomposition products: Acidic and irritating fumes and oxides of carbon, when heated to decomposition.

Section 11: Toxicological Information

For Methanol (Methyl alcohol):

Acute toxicity:

Oral rabbit LD50: 14200 mg/kg

Inhalation mouse LC50: 41000 ppm (6 h)

Other exposure effect:

On the Skin: May cause irritation.

On the Eye: May cause irritation.

Sensitization: No sensitizing effects were known.

Additional toxicological information: To the best of our knowledge, the acute and chronic toxicity of this substance is not fully known. Toxic to the central nervous system and

optic nerve; may cause blindness. Danger through skin absorption. No classification data on the carcinogenic properties of this material is available from NTP, IARC, or OSHA.

For Nitric Acid:

Acute toxicity:

Inhalation rat LC50: 0.13 mg/L (4 h)

Oral (human) LDLo: 430 mg/kg.

Other exposure effect:

On the Skin: Strong corrosive effect.

On the Eye: Strong corrosive effect.

Sensitization: No sensitizing effects known.

Additional toxicological information: To the best of our knowledge, the acute and chronic toxicity of this substance is not fully known. No classification data on the carcinogenic properties of this material is available from NTP, IARC or OSHA.

Section 12: Ecological Information

Toxicity:

Aquatic toxicity: No information.

Persistence and degradability: No information.

Behavior in environmental system:

Bioaccumulative potential: No information.

Mobility in soil: No information.

Additional ecological information: No information.

Other adverse effects: No information.

Section 13: Disposal Considerations

Place in a chemical waste container for proper disposal in an approved waste disposal facility. Dispose of the content and container in accordance with local, regional, national, and international regulations.

Section 14: Transport Information

D.O.T. shipping name: Flammable liquid, corrosive, n.o.s., (Methanol, Nitric Acid)

D.O.T. Hazard class: 3, 8

UN number: UN2924

Packing group: II

Section 15: Regulatory Information

Not meant to be all-inclusive, selected regulations represented

OSHA status: These items meet the OSHA Hazard Communication Standard (29 CFR 1910.1200) definition of a hazardous material.

TSCA status: All components are listed.

Section 16: Other Information

Disclaimer: The information above is believed to be accurate and represents the best information currently available to us.

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